

PROJECT 04

Active Directory User and Group Administration

Windows Server 2025 | Identity Lifecycle | AGDLP | Least Privilege

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ENVIRONMENT	Personal Windows Enterprise Lab
VERSION	v1.0.0
RELEASE DATE	August 11, 2026
STATUS	Complete

Executive Summary

Implemented an Active Directory identity-management foundation for **corp.mslab.test**. The project created and classified departmental users, built an AGDLP security-group model, separated administrative credentials, staged a disabled service identity, and executed a controlled offboarding workflow with PowerShell validation.

Business Problem

Enterprise directories require repeatable account provisioning, role-based access, privilege separation, and traceable offboarding. Assigning permissions directly to users creates administrative debt and weakens auditability. This lab establishes reusable identity controls before domain clients and file-server permissions are introduced.

Project Objectives

- Create eight users in department-specific OUs and populate classification attributes.
- Implement Accounts - Global groups - Domain Local groups - Permissions (AGDLP).
- Create a separate administrative identity protected from delegation.
- Create a disabled, unprivileged service-account placeholder.
- Disable, deprovision, document, and relocate a stale user without deleting evidence.
- Validate account state, group membership, nesting, and OU placement with PowerShell.

RESULT

All planned controls passed. The permissions layer is intentionally reserved for the CORP-SRV01 file-server project.

Environment

Component	Implementation
Forest	corp.mslab.test
Domain controller	CORP-DC1 Windows Server 2025
Management tools	ADUC and Active Directory PowerShell module
Rollback control	P04-Pre-Identity-Objects checkpoint

Identity and Group Standards

Consistent names communicate object purpose and reduce mistakes during administration and auditing.

Object	Convention	Examples
Standard user	First initial + surname	msalazar, ahart
Administrative user	adm_ prefix	adm_msalazar
Service identity	svc_ prefix	svc_labbackup
Global security group	GG_ prefix	GG_Finance_Users
Domain-local resource group	DL_ prefix	DL_FS_Finance_RW

Departmental Accounts

Department	Accounts	OU
Finance	ahart, vchen	Corp / Users / Finance
Human Resources	nbrooks, lriviera	Corp / Users / Human Resources
Information Technology	msalazar, pshah	Corp / Users / Information Technology
Operations	tmorgan, mreed	Corp / Users / Operations

Control Design

Standard accounts perform normal work. Administrative tasks use a separate protected identity. Departmental membership is assigned through global groups; resource access is represented by domain-local groups. Stale identities are disabled, stripped of access, documented, and retained in a dedicated OU for review.

SECURITY

The design minimizes standing privilege, preserves accountability, and prepares clean permission assignments for the later file-server project.

Change Control and User Validation

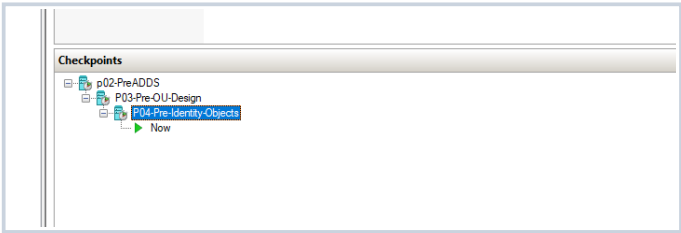


Figure 1 - Rollback checkpoint

P04-Pre-Identity-Objects provides a lab rollback point before directory changes.

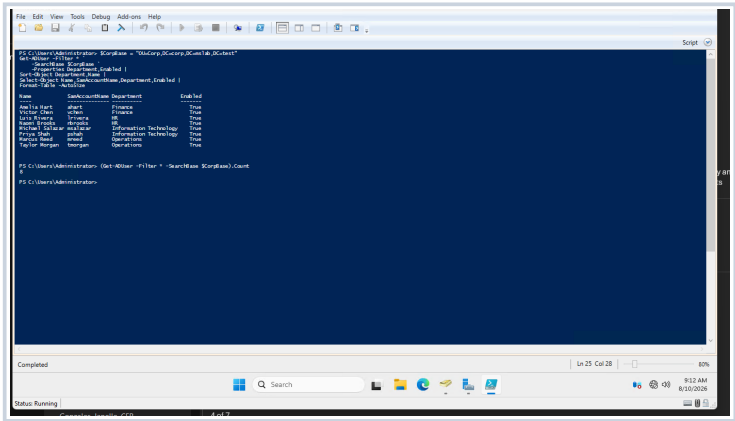


Figure 2 - Eight-user validation

PowerShell confirms account names, departments, enabled status, and a total count of eight.

The checkpoint documents controlled change preparation. User validation verifies that each planned identity exists beneath the Corp OU, is enabled at provisioning, and contains the intended Department attribute.

PASS

8 planned user accounts created, classified, enabled, and validated.

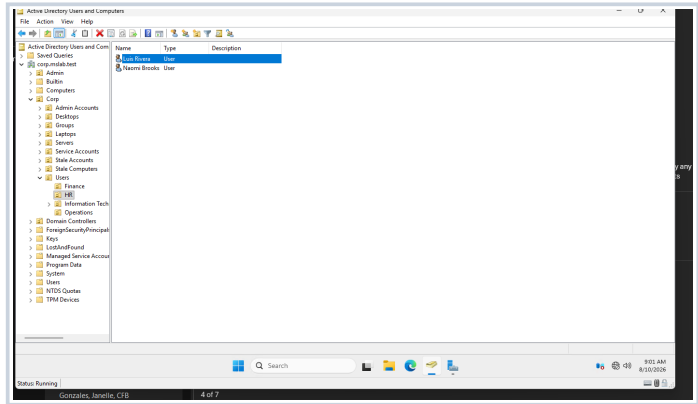


Figure 4 - Human Resources OU
Naomi Brooks and Luis Rivera are placed in Human Resources.

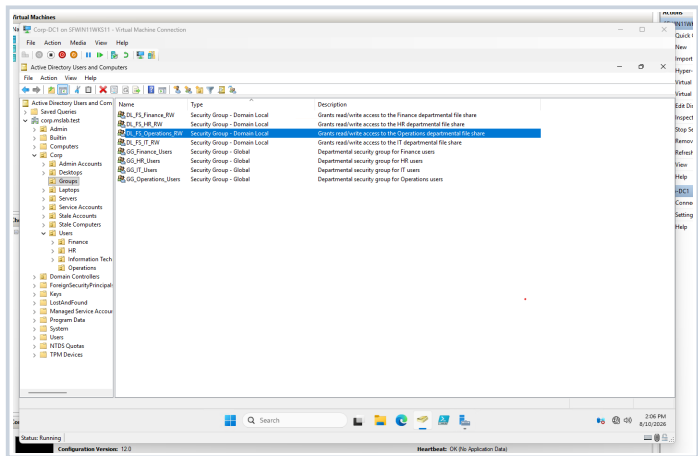
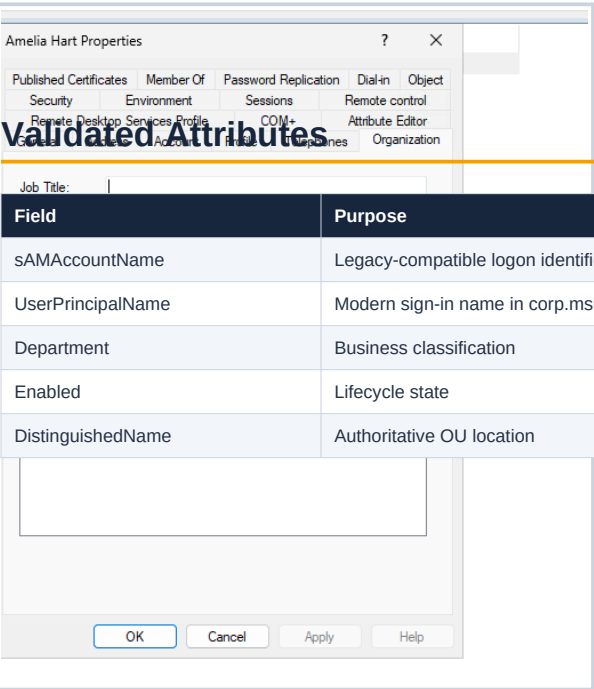


Figure 6 - Operations and group structure
Departmental identities and security groups are visible in ADUC.

Account Classification



OU placement and the **Department** attribute serve different purposes. The OU controls administrative and policy scope; the attribute supports search, reporting, identity classification, and future automation.

Figure 7 - Department attribute

The Organization tab records the user department separately from OU placement.

AGDLP Group Objects

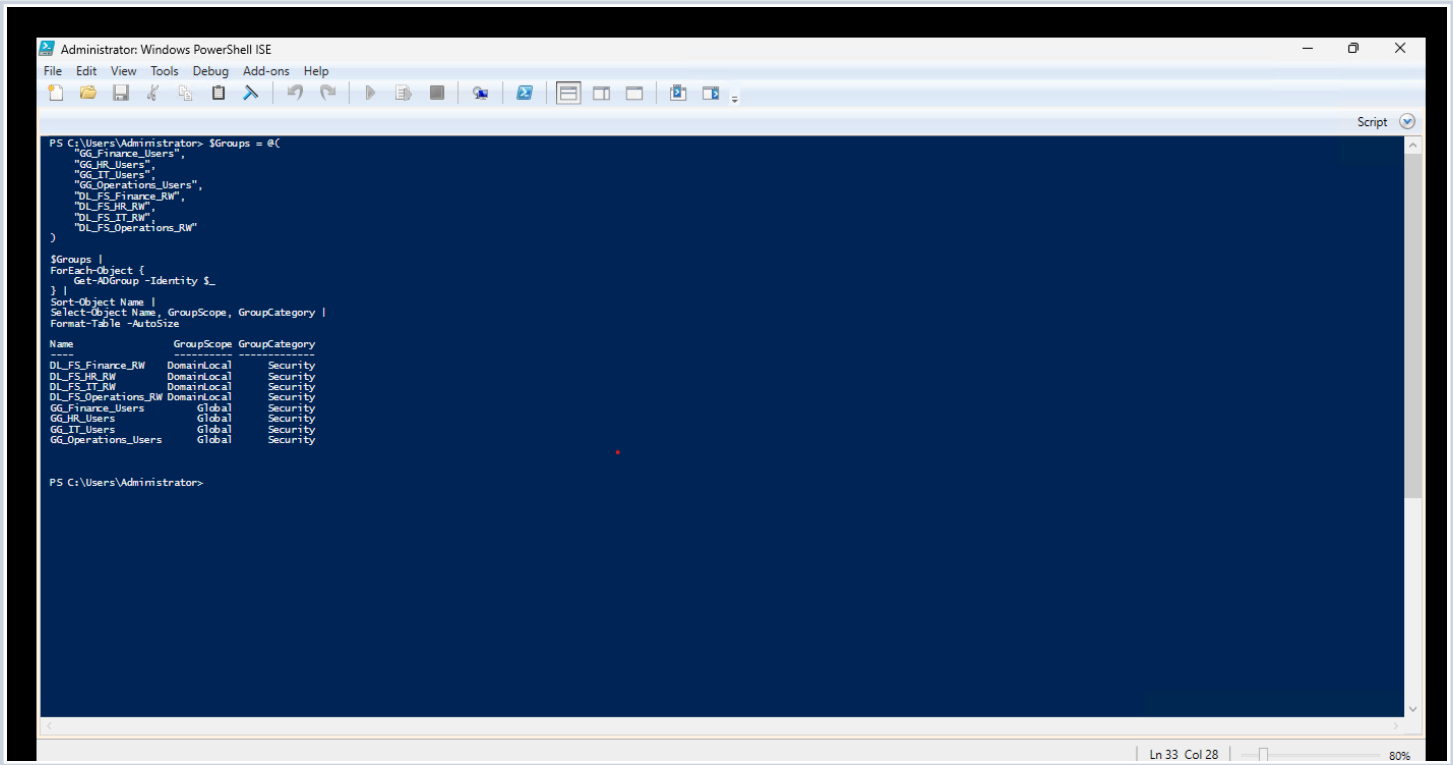


Figure 8 - Group scope and category validation
PowerShell confirms four Global/Security groups and four DomainLocal/Security groups.

Global account groups	Domain-local resource groups
GG_Finance_Users	DL_FS_Finance_RW
GG_HR_Users	DL_FS_HR_RW
GG_IT_Users	DL_FS_IT_RW
GG_Operations_Users	DL_FS_Operations_RW

MODEL

Accounts are added to Global groups; Global groups are nested in Domain Local groups; permissions will be assigned to the Domain Local groups.

Departmental Group Memberships

```
PS C:\Users\Administrator> $GlobalGroups = @(
    "GG_Finance_Users",
    "GG_HR_Users",
    "GG_IT_Users",
    "GG_Operations_Users"
)
```

Validated membership:

Group	Users
GG_Finance_Users	ahart, vchen
GG_HR_Users	nbrooks, lrivera
GG_IT_Users	msalazar, pshah
GG_Operations_Users	tmorgan, mreed (before offboarding)

PASS

No user was added directly to a domain-local resource group.

```
Name                SamAccountName ObjectClass
-----
Michael Salazar    msalazar       user
Priya Shah         pshah          user

=== GG_Operations_Users ===
Name                SamAccountName ObjectClass
-----
Taylor Morgan      tmorgan        user
Marcus Reed        mreed          user

PS C:\Users\Administrator>
```

Figure 9 - Global-group membership validation
Each departmental global group contains exactly its two assigned users.

AGDLP Nesting Validation

```
PS C:\Users\Administrator> $DomainLocalGroups = @(
    "DL_FS_Finance_RW",

```

The implemented path is:

ACCESS User -> GG_Department_Users -> DL_FS_Department_RW -> folder permission (future project)

```
Get-ADGroupMember -Identity $GroupName |
    Select-Object Name, SamAccountName, ObjectClass |
    Format-Table -AutoSize |
    Out-Host
}

=== DL_FS_Finance_RW ===
Name                SamAccountName      ObjectClass
----                -
GG_Finance_Users    GG_Finance_Users    group

=== DL_FS_HR_RW ===
Name                SamAccountName      ObjectClass
----                -
GG_HR_Users         GG_HR_Users         group

=== DL_FS_IT_RW ===
Name                SamAccountName      ObjectClass
----                -
GG_IT_Users         GG_IT_Users         group

=== DL_FS_Operations_RW ===
Name                SamAccountName      ObjectClass
----                -
GG_Operations_Users GG_Operations_Users group

PS C:\Users\Administrator>
```

This structure allows personnel changes to be handled through department membership while resource permissions remain stable. It also creates a clear audit trail between identities, roles, and resources.

Figure 10 - Domain-local nesting

Each DL_FS group contains only its matching GG departmental group.

Privileged Identity and Least Privilege

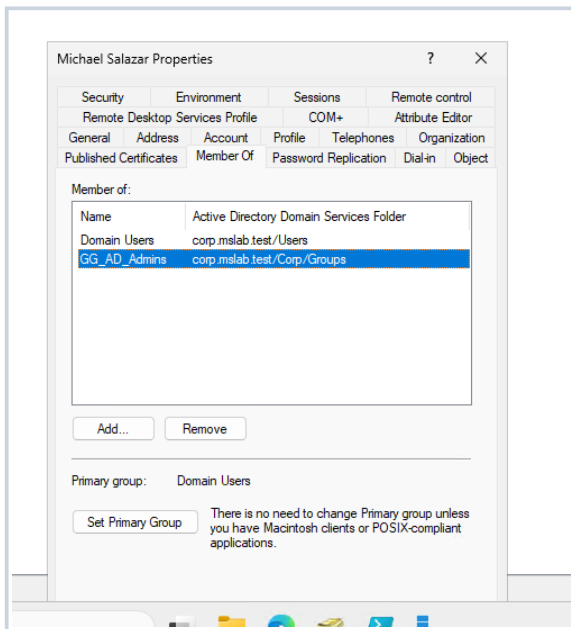


Figure 11 - Administrative group membership
The separate adm_msalaraz identity belongs to GG_AD_Admins.

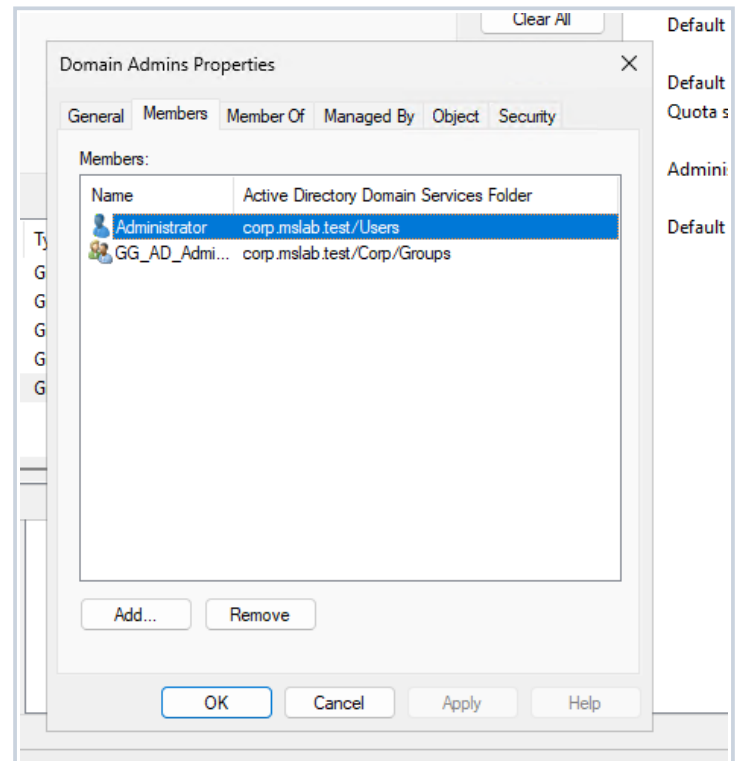


Figure 12 - Nested Domain Admin access
GG_AD_Admins is nested in the built-in Domain Admins group.

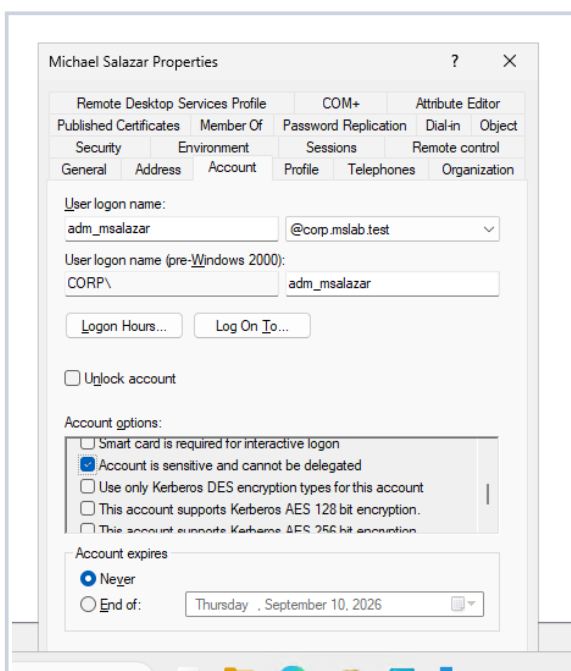


Figure 13 - Delegation protection
Account is sensitive and cannot be delegated is enabled.

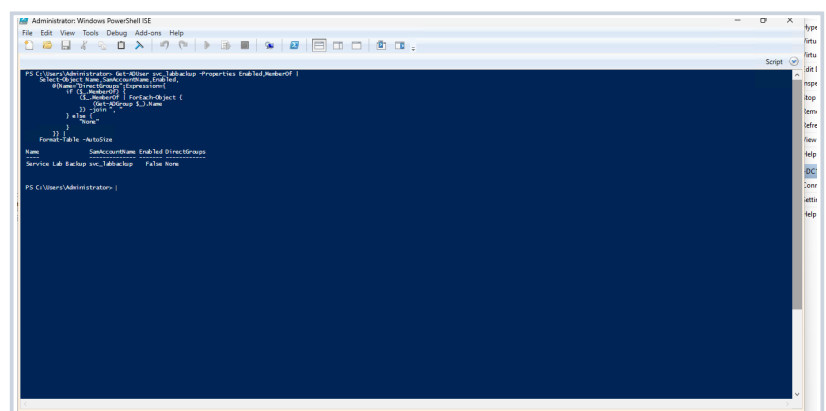


Figure 14 - PowerShell validation
The account is enabled, AccountNotDelegated is True, and both membership layers validate.

Service Identity Control

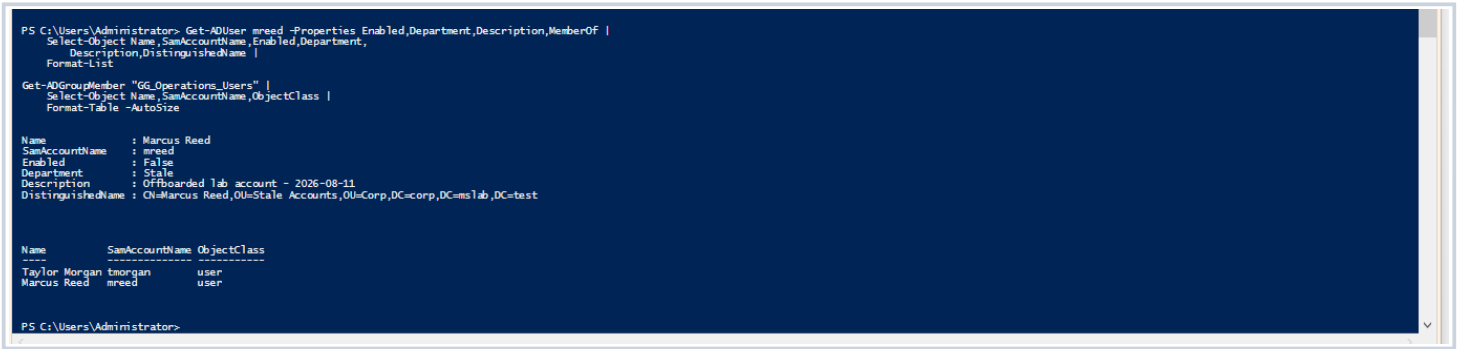


Figure 15 - Service-account validation
svc_labbackup is disabled and has no direct group memberships.

The account is a deliberately disabled placeholder for a future backup service. It has no standing access and no privileged memberships. When CORP-SRV01 joins the domain, the intended service role will be implemented with a group Managed Service Account so Windows can manage its password automatically.

PASS

Enabled = False | DirectGroups = None

Risk Treatment

Leaving the placeholder disabled prevents accidental use before the service, host, rights, and monitoring requirements are defined. The later gMSA implementation will remove the need for a manually managed service password.

Stale-Account and Offboarding Workflow

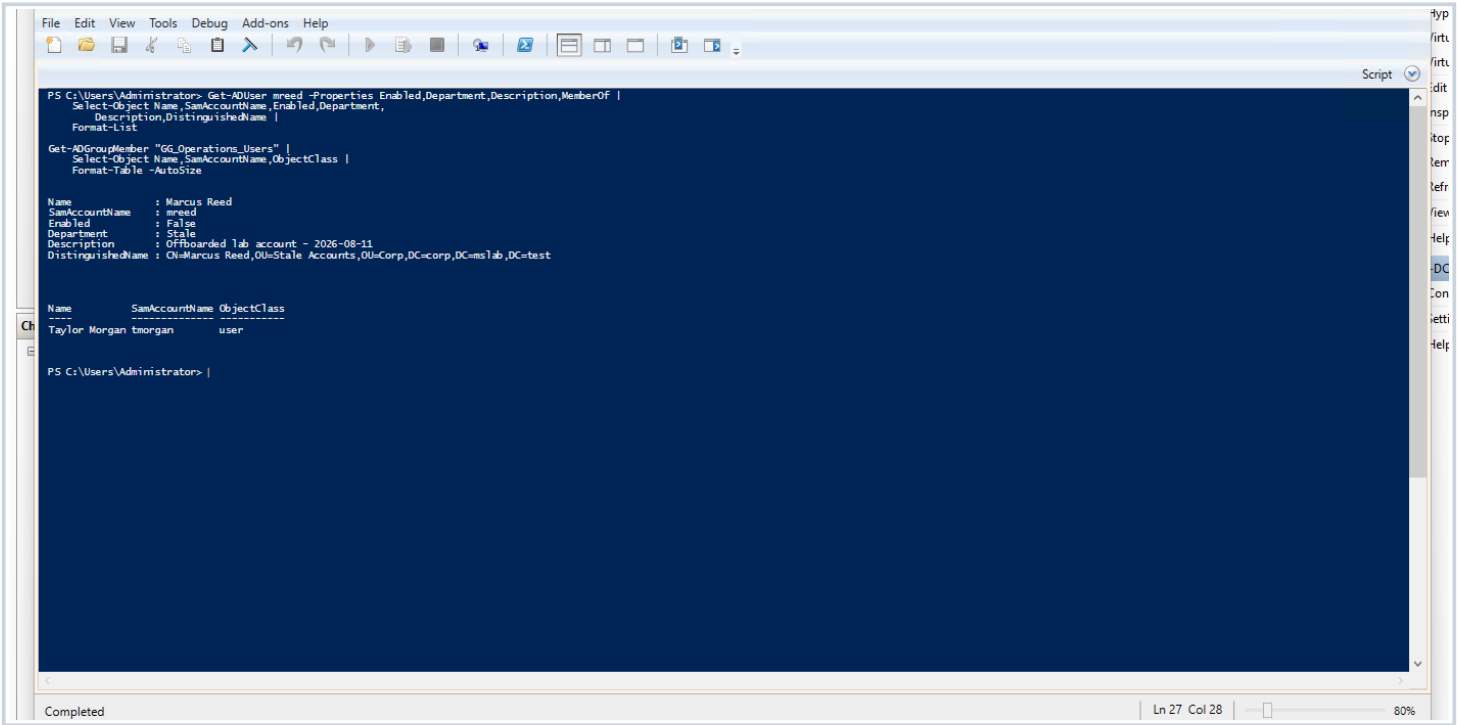


Figure 16 - Final offboarding validation
Marcus Reed is disabled, documented, moved to Stale Accounts, and removed from GG_Operations_Users; Taylor Morgan remains.

Control	Validated result
Disable identity	mreed Enabled = False
Preserve history	Department remains Operations
Document action	Offboarded lab account - 2026-08-11
Quarantine object	DN includes OU=Stale Accounts,OU=Corp
Remove access	GG_Operations_Users contains only tmorgan
Retention decision	Account retained; not deleted

PASS

The account lifecycle moved cleanly from active employment to disabled, access-removed, documented stale status.

Validation Matrix

Control	Status	Evidence result
Rollback checkpoint	PASS	P04 checkpoint present
User provisioning	PASS	8 users; correct departments and enabled state
OU placement	PASS	Four department-specific OUs
Group definitions	PASS	4 Global + 4 Domain Local security groups
Global membership	PASS	Two users per department before offboarding
AGDLP nesting	PASS	One matching GG in each DL group
Privileged identity	PASS	Separate account; protected delegation; nested admin access
Service identity	PASS	Disabled; no direct groups
Offboarding	PASS	Disabled, moved, documented, access removed

Skills Demonstrated

Active Directory Users and Computers; Active Directory PowerShell; user provisioning; identity attributes; OU administration; security-group scope; AGDLP; nested authorization; privileged-account separation; delegation protection; service-account risk control; offboarding; evidence-based validation.

OUTCOME

Project 04 is complete. The next infrastructure dependency is domain joining CORP-SRV01 and MGMT-CL01 before resource permissions and managed service identities are implemented.

Portfolio and Interview Handoff

Resume Bullet

- Built an Active Directory identity-management lab for a Windows Server 2025 domain, provisioning and classifying departmental users, implementing an AGDLP authorization model, separating privileged credentials, protecting administrative delegation, and validating a controlled offboarding process with PowerShell.

STAR Interview Story

STAR	Response
Situation	The new enterprise lab had a functioning forest and OU hierarchy but no standardized users, roles, or lifecycle controls.
Task	Create an auditable identity foundation that supports future file-server permissions and system administration.
Action	Provisioned eight departmental users, populated business attributes, created Global and Domain Local security groups, implemented AGDLP nesting, separated an administrative identity, protected it from delegation, staged a disabled service identity, and offboarded a test user through a documented stale-account workflow.
Result	All controls passed PowerShell validation. The environment now has a scalable identity and access model ready for domain-joined servers, clients, and resource permissions.

Lessons and Next Steps

OU location and Department attributes are independent and must be managed intentionally. PowerShell formatting can buffer objects, so Out-Host was used to display each membership set beneath its correct heading. The next project will domain-join CORP-SRV01 and MGMT-CL01, followed by file services, NTFS/share permissions, and the final AGDLP permissions layer.

Version History

Version	Date	Change
v1.0.0	August 11, 2026	Initial completed release with user, AGDLP, privileged/service identity, and offboarding evidence.